Mohamed Alaa Abdelhamed

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Professional Summary

Results-driven Computer Engineering student with a robust portfolio in Federated Learning, Embedded IoT, and Full-Stack Development. Proven expertise in designing scalable AI architectures (AuraVIT), hardware-software integration (ESP32/Verilog), and data-driven applications. Experienced in cybersecurity and automation, eager to solve complex engineering challenges.

Education

Nile University Sept 2021 – Jan 2026

Bachelor of Engineering (B.E.) in Electrical & Computer Engineering *GPA: 3.7*

- Honors: Dean's Honors List (Nov 2023)

Work Experience

Railway Engineering Intern Aug 2024 – Sep 2024

Alstom Mobility

- Acquired hands-on experience with Automatic Train Control (ATC) and Automatic Train Supervision (ATS) for railway automation.
- Assisted in train maintenance, diagnostics, and mechanical repairs.

Network Administration & Cybersecurity Trainee Jul 2024 – Aug 2024

Ministry of Information and Communication Technology

- Gained knowledge in network security, data center management, and cybersecurity best practices.
- Worked on network troubleshooting and security protocols.

Undergraduate Teaching Assistant Nov 2022 – May 2024

Nile University

- Programming: Assisted students in C++ and Python problem-solving.
- Electric Circuits: Guided students in building and analyzing electrical circuits.
- Engineering Drawing:Delivered instructional content on engineering design principles.

MATLAB Instructor Mar 2023 – Jun 2023

Nile University

- Designed and delivered introductory MATLAB workshops.

Projects

AuraVIT: A Resource-Efficient 2D Hybrid Transformer Framework for Federated Lung Tumor Segmentation May

- Developed a FL framework for lung tumor segmentation using custom **Lightweight AuraViT** models.
- Implemented **Federated Ensemble** and **Transfer Learning** strategies to handle non-IID medical data.
- Integrated Client Drift Detection and a **Federated Bagging** mechanism to enhance model generalization.

ESP32 AI Environmental Monitor (IoT + GenAI) Nov 2025 – Dec 2025

- Engineered a safety system using ESP32, MQ135 (Air Quality), and Flame sensors with MQTT/Blynk integration.
- Built a Python **Edge Gateway** utilizing Google's **Gemini API** to analyze sensor data and generate human-readable safety reports.

F1 Predictor : ML Race Analytics Oct 2025 – Dec 2025

- Built a modular prediction engine using **Gradient Boosting Regressors** and **FastF1** API for telemetry data.
- Implemented track-specific training pipelines to capture circuit characteristics, optimizing for race outcome accuracy.

MIPS Single-Cycle Processor with DSP Unit Feb 2025 – May 2025

- Designed and simulated a MIPS CPU in **Verilog** (Xilinx Vivado), extended with a custom **DSP Unit**.
- Optimized architecture for efficient digital signal processing operations compared to standard ALU instructions.

Federated Learning for Network Intrusion Detection Jul 2025 – Aug 2025

- Conducted a comparative study of FL vs. Centralized Learning for NIDS using UNSW-NB15 datasets.
- Evaluated DNN, LSTM, and Random Forest models, achieving competitive accuracy with privacy preservation.

Electric Car for EVER Competition Feb 2024 – Aug 2024

- Contributed to manufacturing and control system design.

Technical Skills

Languages: Python, C++, C, Verilog
AI/ML: PyTorch, Scikit-Learn, Federated Learning, TensorFlow, OpenCV, Pandas
Embedded & Hardware: ESP32 (ESP-IDF), Arduino, FPGA (Vivado), MQTT, PCB Design
Tools & Frameworks: MATLAB, Git, FastF1, Gemini API
Soft Skills: Technical Writing, Project Management, Agile Methodology, Problem-Solving

Certifications

Cisco Certified Network Associate (CCNA) – Cisco Networking Academy

Linux Essentials – Cisco Networking Academy